

Technology News

A robotic hand, controlled via mind

For the first time, a patient has been able to make complex movements, using his mind, to control a robotic hand connected to his nervous system. Pierpaolo Petruzzello, a 26-year old Italian, lost his left hand and forearm in a car accident last year. During an experiment which lasted a month, an artificial hand, controlled by his thoughts, was connected to the stump of his arm by electrodes.

The prosthetic was not implanted on the patient but only connected through the electrodes. During the month when the electrodes were being connected, Petruzzello learned to wiggle the robotic fingers independently, make a fist, grab objects and make other movements.

After Petruzzello recovered from the microsurgery, it only took him a few days to master the use of his ro-

botic hand. By the time the experiment was over, the hand obeyed the commands it received from his brain, in 95 per cent of cases.

Experts believe that the experiment is an important step forward in creating an interface between the nervous system and prosthetic limbs. But ensuring that such a system remains functional inside the human body for years and not just a month, is a challenge now.

A robot that reads human emotions

Gone are the days when robots were limited to being automated machines complying to just given orders. Now European researchers in robotics, psychology and cognitive sciences have developed a robot that can predict the intentions of its human partner and act more like a companion than a machine.

Under the European Union-funded JAST project, researchers have made some significant discoveries about the cognitive processes involved in joint action and decision-making. Most importantly, they scrutinised the ways in which observation plays an important part in joint activity.

Scientists have already shown that a set of 'mirror neurons' is activated when people observe an activity. These neurons resonate as if they were mimicking the activity; the brain learns about an activity by effectively copying what is going on. In the JAST project, a similar resonance was discovered during joint tasks: people observe their partners and the brain copies their action to try and make sense out of it.

ROAMS robot makes 3D maps on the move

A new robotics vehicle called ROAMS, or remotely operated and autonomous mapping system, uses various existing mapping technologies such as a rotating laser and video camera to generate 3D coloured maps of its surroundings.

It employs a mirror-based LIDAR (light detection and ranging) system that bounces a laser off a rapidly rotating mirror and gleans environmental information from how long it takes for each pulse to bounce back. The system takes about 30 seconds to scan a 160-metre-wide area. A colour camera on the rotating frame provides colour information that is added to the map later on. The same resolution can also be maintained by automatically adjusting the scanning process depending on the proximity of objects.

Such a vehicle could be used by military personnel for low-cost architectural surveying and map making. It was created by researchers at the Stevens Institute of Technology in Hoboken, New Jersey, with funding from the US army.



Petruzzello took part in an experiment in which a group of European scientists say they successfully connected him to the robotic hand, using electrodes to his body, allowing him to control the prosthetic with his thoughts and feel sensations in the artificial limb